



Illustration of Wi-Fi 7, highlighting its key features

### Tips for overcoming Wi-Fi 6 and Wi-Fi 7 design challenges

In the evolving wireless communication landscape, optimising RF system performance is essential as Wi-Fi 6 and 7 push the boundaries of high-speed, low-latency connectivity.

- **Focus on RF optimisation:** Ensure that your design includes high-quality RF components, such as low-noise amplifiers (LNAs), RF switches, and power amplifiers (PAs), that can support high-bandwidth communication and higher QAM levels.
- **Leverage power-saving features:** Use Wi-Fi 6 and 7's built-in power-saving features, such as target wake time (TWT), to optimise battery life, particularly in IoT and mobile applications.
- **Invest in antenna diversity:** Implement multi-band antenna designs that are optimised for 2.4GHz, 5GHz, and 6GHz operation, while ensuring minimal interference between bands.
- **Enhance thermal management:** Consider integrating thermal sensors and dynamic power management features to monitor device temperature and prevent overheating in high-performance applications.
- **Use smart network management:** Implement advanced network scheduling algorithms to minimise latency, especially for real-time applications like AR/VR or gaming. Ensuring proper QoS handling will help prioritise critical data streams.

akin to having two pipes for data flow instead of one, which allows for faster data transmission and reduced latency. If one band becomes congested, the device can automatically switch to the other, ensuring seamless communication without delays. MLO also enhances reliability, as devices can switch between bands without interruption in data flow, providing a seamless experience in real-time applications, such as online gaming or video conferencing.

**Latency reduction:** Wi-Fi 7 specifically addresses applications that require ultra-low latency, such as AR/VR, machine learning, and real-time data blending. By utilising MLO, devices can transmit and receive data faster and more reliably, significantly reducing delays

that could hinder performance in latency-sensitive environments.

**Enhanced throughput:** With the ability to use both frequency bands simultaneously, Wi-Fi 7 can potentially double the data rates, catering to applications requiring high bandwidth, such as 8K streaming or cloud gaming.

**Low-latency performance for AR/VR and real-time applications:** Wi-Fi 7 is designed to reduce latency dramatically, making it ideal for emerging real-time applications, such as augmented reality (AR), virtual reality (VR), and cloud gaming. With sub-millisecond latency goals, Wi-Fi 7 aims to deliver ultra-responsive, lag-free connectivity, creating immersive experiences for users.

**Increased network efficiency with coordinated OFDMA:** Wi-Fi 7 will enhance OFDMA with more efficient coordinated multi-user resource allocation. This will allow for even more precise channel allocation, improving performance in environments with many active devices. As the number of connected devices continues to grow, this feature will ensure that Wi-Fi networks maintain high performance without bottlenecks.

### Challenges and future adoption

Adopting Wi-Fi 7 will present both opportunities and challenges for organisations. The good news is that Wi-Fi 7 is backward compatible with earlier versions, meaning businesses can gradually upgrade their infrastructure without replacing every device. This flexibility will ease the transition, particularly for industries already heavily invested in Wi-Fi 6 networks.

However, the key challenge will be cost. Wi-Fi 7's advanced features, such as MLO and ultra-low latency, will come at a premium, making it a more suitable option for high-end applications. Organisations will need to carefully evaluate their specific needs and decide whether Wi-Fi 6's efficiency is sufficient or if Wi-Fi 7's low-latency benefits justify the investment.

**Increasing complexity in RF design:** As Wi-Fi standards advance with high-order modulation like 4096-QAM, maintaining precise RF performance is crucial due to its sensitivity to noise, phase shifts, and interference. This necessitates improved signal integrity, requiring highly linear amplifiers and advanced filtering. These needs are addressed with high-linearity power amplifiers (PAs) and low-noise amplifiers (LNAs), designed to minimise distortion, and enhance signal quality, even at high data rates. These components are essential for supporting advanced modulation and MLO, ensuring optimal Wi-Fi performance.